

Inference-Time Scaffolding Across Two Axes: Closing a Cross-Generation Gap and Climbing a Model-Tier Ladder with Rubric-Anchored Adversarial Refinement

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Abstract—Frontier language-model upgrades matter along two practical axes: a direct successor can replace an older flagship, and a cheaper tier can displace a frontier tier if better inference-time scaffolding recovers enough quality. We study both with Rubric-Anchored Adversarial Refinement (RAAR), a five-stage loop that fixes a binary success rubric before generation, diversifies candidates, verifies each adversarially in a fresh context, fuses verified strengths, and exits when a fresh verifier fails to refute the answer or a bounded revision budget is spent. Study 1 compares Claude Opus 4.8+RAAR against Claude Fable 5 single-pass on LvUP-Bench v1. The measured objective gap concentrates in one precision-writing task; there RAAR recovers 5 of the 7 constraint checks the predecessor misses (71%, $n=4$), lifting the macro-average from 0.971 to 0.992 against the successor’s 1.000, with no loop run scoring below the base model in 14 runs. Study 2 introduces LvUP-Bench v2, a headroom-seeking ladder benchmark with 12 objective tasks and 3 judged tasks; one constrained writing item is excluded from primary analysis because collection failed and left an incomplete no-answer cell. Post-hoc, Fable remains near ceiling on most scored objective tasks, so frontier-gap claims use an explicit Fable-over-base headroom filter. On the 5 common verification-shaped tasks where Sonnet trails Fable by more than 0.05 under that documented rule, Sonnet+RAAR closes 0.733 of the filtered gap (95% CI [0.300, 1.067], $n=5$); this headline is threshold-sensitive (0.50–0.73 across tiny-denominator headroom cutoffs 0.02–0.10, peaking at the 0.05 cutoff, and 0.42 pooled across all 10 tasks), and across all 10 common scored tasks Sonnet+RAAR remains 0.137 below Fable single-pass ($p=0.127$). Haiku+RAAR gains 0.274 absolute on 11 matched primary tasks but closes only 0.304 of its Fable gap (95% CI [−0.111, 0.639]). At Sonnet tier, Self-Refine is numerically the strongest loop arm, and on the matched 11-task primary set the loop family lifts Sonnet base from 0.730 to 0.820–0.901 (the upper value is Self-Refine on the same 11-task denominator; its scored-available 12-task macro is 0.873) with no measurable separation among loop variants. A direct-answer Sonnet control falls to 0.478, consistent with deliberation being load-bearing. Across the two studies, the practical reading is bounded but useful: one tier down plus RAAR can narrow the gap toward unscaffolded frontier output (Fable single-pass) on a filtered verification-shaped subset — we did not complete an equal-scaffolding (Fable+RAAR) frontier control — whereas opened coherence tasks and two-tier drops do not recover.

Index Terms—Large language models, test-time compute, inference scaling, self-refinement, LLM-as-a-judge, verification.

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I. Introduction

FRONTIER language models now improve along two practical axes at once. A direct successor replaces an older flagship; below that, cheaper serving tiers shadow the frontier closely enough that a better calling pattern may be more attractive than a weight switch. Both cases raise the same engineering question: how much of the quality difference is recoverable at inference time, by spending more compute through a better scaffold rather than by changing the model? The question matters wherever model swaps are slow or impossible — regulated deployments pinned to audited versions, cost-tiered products, and the long tail of systems that cannot re-validate against every new release. It is also a clean scientific probe: inside one vendor family, gap closure isolates capability progress from the vendor- and data-related confounds that plague cross-family comparisons.

The test-time-compute literature gives a partial answer. With a reliable external verifier, inference compute substitutes for parameters: a compute-optimally sampled small model can outperform a $14\times$ larger one on MATH [1], repeated sampling lifts an inexpensive open code model from 15.9% to 56% coverage on SWE-bench Lite [2], and a trained verifier is worth a $30\times$ parameter increase on GSM8K [3]. But these results live where ground truth is checkable. The negative results live everywhere else: without external feedback, intrinsic self-correction degrades reasoning [4], GPT-4 self-critique collapses on planning tasks [5], self-refinement stalls once the model is confident [6], and models systematically prefer their own generations [7], [8]. Worse, the self-improvement gap saturates within a few iterations [9], and imperfect verifiers impose accuracy ceilings that no resampling budget breaks [10].

This paper studies both axes with one loop, Rubric-Anchored Adversarial Refinement (RAAR). Study 1 revisits the hardest same-family question: can an older flagship, Claude Opus 4.8, recover a direct successor, Claude Fable 5, at inference time? Study 2 asks the deployment question practitioners often face first: how far up a model-tier ladder can the same loop lift cheaper tiers on a benchmark with calibrated headroom? RAAR uses the model’s own weights in five role-separated, fresh-context

stages: rubric externalization before any solution exists, strategy-diversified parallel generation, evidence-quoting adversarial verification, defect-aware fusion, and a burden-of-proof-flipped exit gate that terminates only when a fresh adversarial verifier fails to refute the candidate. The design is deliberately conservative about every published failure mode: no role ever critiques inside the context that generated the artifact (defusing self-preference, whose mechanism is self-recognition [7]), critique is anchored to externally fixed binary rubrics (converting unreliable holistic judgment into reliable per-item checks [11], [12]), and revision rounds are capped at the saturation horizon the self-improvement literature predicts [9].

The two studies land in different but compatible places. On LvUP-Bench v1, the objective generation gap between Claude Opus 4.8 and Claude Fable 5 concentrates in a single precision-writing task, and there RAAR recovers 5 of the 7 constraint checks the predecessor misses (71%, $n=4$), lifting the macro-average from 0.971 to 0.992 against the successor’s 1.000, with no loop run scoring below the base model in any of 14 runs. On LvUP-Bench v2, the overall Sonnet lift is modest on 11 matched primary objective tasks (+0.089, $p=0.521$), but the task-conditional story is sharper: Sonnet+RAAR closes about 73% (threshold-sensitive: 0.50–0.73, or 0.42 pooled) of the filtered Fable gap on the 5 common tasks where Fable’s headroom over Sonnet exceeds 0.05 (95% CI [0.300, 1.067], $n=5$), while Haiku+RAAR improves in absolute score (+0.274, $p=0.042$) yet recovers only 0.304 of its gap (95% CI [−0.111, 0.639]). At Sonnet tier, Self-Refine reaches the best scored-available macro (0.873 over 12 task means) and, on the matched 11-task primary set, the loop family lifts base Sonnet from 0.730 to 0.820–0.901 without a measurable separation among loop variants. Together the studies say that iteration alone is not the active ingredient; the recoverable part of the gap is task-shaped, tier-shaped, and mediated by adversarially gated, rubric-anchored, context-decoupled deliberation.

A scope clarification is important for interpreting the released artifact. The generator comparisons in this version are Claude-family comparisons: Claude Opus 4.8, Sonnet, Haiku, and Claude Fable 5. GPT-5.5 appears only in the blinded judging pipeline. The artifact does not contain a GPT-5.5 generation arm, so this paper does not claim that RAAR has been measured on GPT-5.5 as a generator.

Concretely, the paper does four things. It defines RAAR as a role-separated, rubric-anchored inference scaffold; builds two small benchmark regimes with mechanical checkers and blinded judged tails; collects Claude-family generator outputs under single-pass, looped, and ablation arms; and reports the resulting wins, failures, missing cells, and adversarial audit findings in one inspectable artifact. The goal is not a leaderboard claim. The goal is to make the boundary visible: where call structure appears to recover a model-tier gap, where it fails, and what future controls would be needed for a stronger claim.

Our contributions:

- RAAR, an inference-time loop whose every compo-

nent is motivated by a documented failure mode of naive refinement, and whose key novelty — a generated instance-specific rubric acting as a synthetic weak verifier that gates loop exit on otherwise unverifiable tasks — extends verifier-gated test-time compute beyond math and code (Section III).

- Two benchmark regimes under one artifact: LvUP-Bench v1, a small contamination-controlled cross-generation benchmark, and LvUP-Bench v2, a headroom-seeking ladder benchmark with 12 objective tasks and 3 judged tasks, transparent failed cells, one disclosed task exclusion from primary analysis due to failed no-answer collection, and an explicit post-hoc saturation caveat for frontier-tier cells (Sections IV-A and V).
- A two-study empirical answer: Study 1 evaluates predecessor + loop vs. direct successor; Study 2 evaluates the same loop across cheaper model tiers, with explicit permutation tests for all significance claims and a cross-family judged panel for open-ended tasks (Sections IV and V).
- A practical playbook, not just a win/loss headline: RAAR helps most on verification-shaped tasks with remaining headroom, can narrow a one-tier-down model’s gap to unscaffolded frontier output on the measured headroom slice, does not clearly recover two-tier drops, and remains weak on coherence-shaped prose (Section V-F).
- A diagnostic artifact for follow-on work: the release keeps raw outputs, sanitized Study 1 role-separated message transcripts, scored rows, failed/no-answer cells, judge manifests, checker hardening notes, and adversarial-review findings together so that later work can reproduce, criticize, extend, or falsify the boundary map rather than trust a compressed headline.

II. Related Work

Test-time compute scaling. Snell et al. [1] showed compute-optimal inference scaling can beat a $14\times$ larger model FLOPs-matched; Brown et al. [2] established log-linear coverage growth under repeated sampling; Liu et al. [13] pushed a 1B model past a 405B model on MATH-500 with process-reward-guided search; s1 [14] reproduced o1-style scaling with budget forcing. The consistent precondition is a verifier: Setlur et al. [15] prove verifier-free test-time scaling is suboptimal by a $\sqrt{H}$ factor, and every weak-beats-strong result above relies on automatic checking. Conversely the limits are documented: best-of- n degrades past $n \approx 400$ through verifier hacking [3], false-positive verifiers impose hard ceilings [10], and longer deliberation can hurt via distractor amplification [16]. RAAR inherits the verifier-gated paradigm but manufactures the verifier — an instance-specific rubric generated before any candidate exists — so the paradigm applies to unverifiable tasks, and caps all iteration far below the hacking and saturation regimes. The closest prior comparisons are same-family but cross-scale: Liu et al. [13] pit a 1B Llama against a 405B sibling using trained process

reward models on math; Snell et al.’s comparisons are likewise intra-family with external verifiers on verifiable tasks. Our setting differs on the three axes that matter for the deployment question: frontier predecessor vs. direct successor at comparable scale, no trained or programmatic verifier available to the loop, and open-ended tasks among the targets.

Self-refinement and its failure modes. Self-Refine [17] and Reflexion [18] popularized iterative self-feedback. Subsequent negative results sharpened the picture: intrinsic self-correction without external signals degrades reasoning [4]; self-verification collapses on planning [5]; refinement stalls at confident wrong answers [6]; the generator–validator gap means the internal judge agrees with the internal generator only 76% of the time even for GPT-4 [19]; and self-improvement capacity saturates in a few rounds with collapsing diversity [9], [20]. RAAR treats this literature as a design specification (Section III-B).

Multi-candidate generation and fusion. Self-consistency [21] and LLM-Blender [22] established that sampling plus rank-or-fuse beats single passes; Mixture-of-Agents [23] beat GPT-4o with open models, while Self-MoA [24] showed single-model sampling diversity often suffices — convenient for our single-model setting. Debate-style multi-agent setups help factuality [25] but do not reliably beat self-consistency at matched cost [26]; we therefore use directive-diversified parallel generation rather than inter-agent dialogue, and fuse with defect evidence instead of ranking alone.

Rubrics and checklists. Instance-specific binary checklists materially improve both evaluation agreement and downstream refinement (TICK/STICK [11]; CheckEval [27]); rubric-based rewards yield the largest gains for the weakest judges [12], which is exactly the property a predecessor-model verifier needs; rubric scaffolds also steer generation when shown up front [28]. Prior work scores or trains with rubrics; RAAR is, to our knowledge, the first to let a self-generated rubric gate loop termination at inference time.

Judging hygiene. LLM judges suffer position bias strong enough to flip rankings [29], verbosity bias [30], and self-preference driven by self-recognition [7], [8]. Our evaluation therefore scores objective tasks purely mechanically, and judged tasks with position-swapped pairwise comparisons across three model families, so no family’s self-preference decides any comparison involving its own output.

Scaffolding as a capability lever. On agentic benchmarks, interface and scaffold design alone moves a fixed model by a factor of three [31], simple scaffolds rival agentic ones [32], and deliberate ensembles of weak verifiers approach frontier-model accuracy [33]. RAAR is the pure-text analog: no tools, no retrieval, no environment — the entire lever is call structure.

III. The RAAR Loop

A. Setting and Notation

Let M_w denote a base model (here Claude Opus 4.8) and M_s a target model (here Claude Fable 5), where M_s dominates M_w in single-pass output quality on a task distribution $\mathcal{T}$. Given a task $x \in \mathcal{T}$, a single-pass arm produces $y = M(x)$ with one inference call. An inference-time scaffold S produces $y = S^{M_w}(x)$ using only calls to M_w — the same weights in every role — and no external information. The question we study is whether a scaffold can close the generation gap $\Delta(x) = q(M_s(x)) - q(M_w(x))$, where q is task quality measured by an oracle (hidden tests, exact answers, mechanical checkers) or by a blind cross-family judge panel.

B. Design Principles

RAAR is built from three observations, each grounded in prior failure analyses of naive refinement loops:

P1 — Verification is easier than generation. For the task families we study, checking a candidate against an explicit criterion (recomputing an arithmetic step, tracing code on a boundary input, counting sentences) requires strictly less capability than producing a flawless artifact in one pass. A model too weak to generate M_s -grade output in one call can still be a reliable detector of the defects that separate its output from M_s ’s.

P2 — Self-evaluation in-context is corrupted by self-consistency. When the same context that produced an answer is asked to critique it, the critique inherits the generation’s blind spots and exhibits sycophantic self-approval; this is the documented failure mode of intrinsic self-correction. The fix is structural, not exhortative: every evaluative role in RAAR runs in a fresh context that never sees the reasoning that produced the artifact under review, and is told the artifact was written by someone else.

P3 — Exit conditions must flip the burden of proof. A loop that exits when a critic approves converges to whatever the critic is too lazy to object to. RAAR exits only when an adversarial verifier, explicitly mandated to hunt for defects with per-criterion evidence, fails to refute the answer. Approval is the absence of surviving objections, not the presence of praise.

C. The Five Stages

Algorithm 1 gives the full loop; all five roles are played by M_w in stateless, fresh-context calls.

Stage 1: Rubric externalization. Before any solution attempt, a fresh instance is asked to produce 8–12 binary, mechanically checkable success criteria for the task: correctness requirements, exact output-format requirements, and the typical failure modes of the task family. The rubric instance is forbidden from solving the task, so that the rubric does not inherit a particular solution’s blind spots — an instruction, not a guarantee: Section VI-B documents one run in which the rubric violated it and the loop survived anyway. The rubric then anchors every

downstream role: generators see it as the grading contract, verifiers check against it criterion by criterion, and the fuser uses it to arbitrate between candidates. This converts the vague instruction “be correct” into a concrete, shared checklist — the loop’s fixed point is defined before the first candidate exists.

Stage 2: Diversified generation. $K=3$ generators run in parallel, each with the task, the rubric, and a distinct strategy directive (decompose-first; enumerate-typical-failures-first; direct-expert-pass with line-by-line re-verification). Distinct directives decorrelate failure modes: the candidates fail in different places, which is what makes cross-candidate fusion (Stage 4) informative.

Stage 3: Adversarial verification. Each candidate is reviewed by a fresh verifier instance that (i) is told the answer “was written by someone else and likely contains defects,” (ii) must walk every rubric criterion quoting evidence, (iii) must independently recompute any computation rather than trusting it, and (iv) must grade each defect FATAL / MAJOR / MINOR. The mandate to hunt — with per-criterion evidence obligations — is what defeats the rubber-stamp equilibrium.

Stage 4: Fusion. A fresh instance receives all K candidates with their defect reports and composes a single answer: start from the strongest candidate, graft superior fragments from the others, fix every flagged defect (re-verifying each flag, since verifiers can be wrong). Fusion-with-evidence is strictly stronger than best-of- K selection: it can repair the best candidate’s verified defects with material from candidates that got those parts right.

Stage 5: Burden-flipped gate. A fresh adversarial verifier reviews the fused answer under the same mandate. If it reports any FATAL or MAJOR defect, the defect report is handed to a fresh revision instance (which never sees the fuser’s reasoning, only its artifact), and the gate repeats; the loop exits when the verifier fails to refute, or after $R=2$ revisions. Note the asymmetry of the second exit: a cap-exhausted run returns its final revision unverified; Section VI-A reports how often this happened and what it cost.

D. Why the Gate is the Kick

Each stage of RAAR has published precedent except the one that does the disproportionate work: the combination, in Stages 3 and 5, of rubric-anchored adversarial verification with a burden-of-proof-flipped exit. We call this the loop’s kick, and the mechanism is worth stating precisely.

A single-pass generation by M_w fails not because M_w lacks the knowledge to avoid its mistakes, but because generation must get every decision right jointly in one sampled trajectory, while the model’s reliability per decision is below the threshold that the conjunction requires. Verification decomposes the conjunction: each rubric criterion is checked independently, in a fresh context, with no commitment to any prior trajectory. As a motivating heuristic — not a validated mechanism — if M_w detects

Algorithm 1 RAAR: Rubric-Anchored Adversarial Refinement

Require: task x , base model M_w , $K=3$, $R=2$

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1:  $\rho \leftarrow M_w(\text{Rubric}(x))$   $\triangleright$  no solving allowed
2: for  $i = 1 \dots K$  in parallel do
3:    $c_i \leftarrow M_w(\text{Gen}(x, \rho, d_i))$   $\triangleright$  directive  $d_i$ 
4: end for
5: for  $i = 1 \dots K$  in parallel do
6:    $v_i \leftarrow M_w(\text{Verify}(x, \rho, c_i))$   $\triangleright$  fresh context
7: end for
8:  $y \leftarrow M_w(\text{Fuse}(x, \rho, c_{1..K}, v_{1..K}))$ 
9: for  $r = 1 \dots R$  do
10:   $g \leftarrow M_w(\text{Verify}(x, \rho, y))$ 
11:  if  $g$  reports no FATAL/MAJOR defect then
12:    break
13:  end if
14:   $y \leftarrow M_w(\text{Revise}(x, \rho, y, g))$ 
15: end for
16: return  $y$ 

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a defect of the relevant class with probability d per inspection, and inspections are roughly decorrelated across fresh contexts, the probability that a defect survives the candidate-stage review plus R gate reviews falls like $(1 - d)^{R+1}$, while a single-pass generation gets exactly one chance to avoid introducing it. Both assumptions can fail: detection rates are defect-class-dependent, and the shared rubric correlates inspections across rounds — our T4 counting residue and the rubric-poisoning run of Section VI-B are exactly such failures, and we report them as the method’s boundary. The heuristic’s core survives the caveats: the loop converts a per-trajectory reliability problem into a per-criterion detection problem, and detection is the easier side of the generator-verifier asymmetry [3], [15], [34].

Three structural choices keep this argument from collapsing in practice, each neutralizing a documented failure mode:

(i) The rubric exists before any candidate. A critic asked “is this good?” free-associates criteria correlated with the answer’s own framing — the generator-validator inconsistency of [19]. Fixing 8–12 binary criteria from the task statement alone makes the verification target exogenous to every candidate, so a candidate cannot look correct merely by defining the terms of its own grading. This is why rubric quality gains are largest for weak judges [12]: the rubric carries the judgment structure the judge would otherwise have to improvise.

(ii) The verifier is adversarial by mandate, with evidence obligations. Free-running critics rubber-stamp once the artifact looks fluent — the confidence stall of [6] and the self-approval documented by [4]. RAAR’s verifier is told the artifact was written by someone else (attacking the self-recognition channel of self-preference [7]), is required to walk every criterion quoting evidence, and is required to recompute any computation it encounters. It may conclude PASS only as a failure of an active refutation attempt,

never as an impression.

(iii) The exit condition is refutation-failure, not approval. A loop that exits on approval optimizes for whatever makes critics approve — Goodhart pressure toward confident-sounding output. A loop that exits only when a fresh adversary finds nothing FATAL or MAJOR optimizes for indefensibility of objections, which is the actual quality target. The asymmetry matters: approval is one bit produced cheaply; a failed hunt is a costly certificate that correlates with genuine absence of detectable defects. Because gap saturation bounds how many rounds can help [9], the gate is capped at $R=2$ revisions — the certificate is bought at bounded cost.

The ablations in Section IV make the attribution concrete: best-of-3 shares RAAR’s diversity but not its verification and gains little; Self-Refine shares its iteration but not its role separation or rubric anchor and gains less, exactly as the negative-results literature predicts [4], [5].

IV. Study 1: Cross-Generation Gap on LvUP-Bench v1

A. LvUP-Bench v1 and Setup

Study 1 is the original cross-generation experiment in this paper, now reframed as the first half of a two-study argument. Comparing scaffolds across models requires unusual care: public benchmark items may sit in either model’s training set, and judge models prefer their own family’s prose. LvUP-Bench v1 controls both.

Objective core (5 tasks). (i) T1 interval algebra: implement an IntervalSet class with open/closed endpoint semantics, set difference (remove) with closedness flips at cut points, point intervals, and canonical string output; graded by 40 hidden tests covering endpoint-merge and removal case analysis. (ii) T2 seeded bug-fix: a sliding-window rate limiter with four seeded boundary/recording bugs and an authoritative prose spec; graded by 10 hidden tests constructed so that partial fixes are caught (one seeded bug masks two others, so only the complete fix passes). (iii–iv) T3 combinatorics ($\times 2$): novel counting problems whose answers (293,749 and 324,288) were computed by two independent programs that agree; graded by exact match. (v) T4 constrained writing: a paragraph that must satisfy 12 simultaneous mechanical constraints; graded by a deterministic checker, score = fraction of constraints met. All objective items were constructed for this study and mechanically verified.

Semi-objective and judged tail. T5 code review is a 60-line database service module with six seeded defects, scored by seeded-defect recall. T6 (multi-region rate-limiter design document) and T7 (WAL/fsync durability explanation) are open-ended and scored by blind pairwise comparison with position swap from a four-judge, three-family panel: GPT-5.5 judge, Gemini, Claude Fable 5, and Claude Opus 4.8. GPT-5.5 is used here only as a judge, not as a generation arm. No family holds a panel majority, and the two Claude judges never disagreed with the cross-family majority in any cell.

Arms and execution. The arms are Claude Opus 4.8 single-pass; RAAR^{Opus 4.8}; Claude Fable 5 single-pass;

TABLE I
Objective-task scores (mean over runs; 1.0 = all hidden tests / constraints / exact answers correct). n per cell: T1, T2 = 2; T3a, T3b = 3; T4 = 4.

task	Opus 4.8				Fable 5
	single	+SR	+Bo3	+RAAR	single
T1 interval code	1.000	1.000	1.000	1.000	1.000
T2 seeded bugfix	1.000	1.000	1.000	1.000	1.000
T3a combinatorics	1.000	0.000	1.000	1.000	1.000
T3b combinatorics	1.000	1.000	1.000	1.000	1.000
T4 constraints	0.854	0.979	0.854	0.958	1.000
macro-average	0.971	0.796	0.971	0.992	1.000

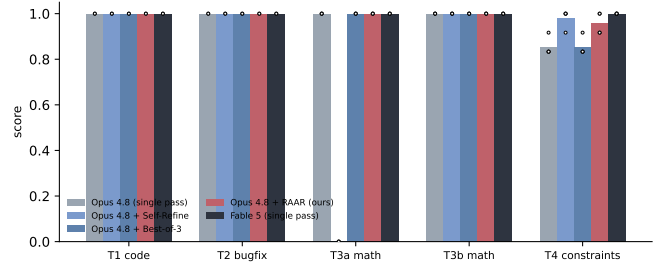


Fig. 1. Per-task scores; dots are individual runs.

Self-Refine on Claude Opus 4.8 with two feedback-refine rounds (5 calls); and best-of-3 on Claude Opus 4.8 with an LLM selector (4 calls). All arms ran on 12 June 2026 against claude-opus-4-8 and claude-fable-5 with vendor-default settings via Claude Code subagent calls. Objective code tasks used 2 independent runs per arm, math tasks 3, the constraint task 4, and judged tasks 1 run per arm. Single-pass arms make 1 call; Self-Refine 5; best-of-3 4; RAAR 9–12 depending on gate outcomes (mean 10.5 over all runs).

Model and tool disclosure. Experimental generation is limited to the model labels recorded in the released run artifacts. Study 1 generation used claude-opus-4-8 and claude-fable-5; Study 2 generator rows in results/v2/all_runs.scored.json are labeled haiku, sonnet, opus, and fable. No GPT-5.5 output is used as a scored generated answer. GPT-5.5 and Gemini-family tools are used as blinded pairwise judges in the released judging artifacts, alongside claude-fable and claude-opus judges in Study 1. The manuscript, code, benchmark, and checks were developed with AI-assisted drafting, coding, debugging, checking, and adversarial review; the author reviewed the artifacts and is responsible for all claims, data, code, and citations.

B. Objective Tasks

Table I and Fig. 1 give per-task mean scores; Fig. 2 the macro-average.

Three findings.

(1) The objective generation gap is narrow and concentrated. On four of five objective tasks, Claude Opus

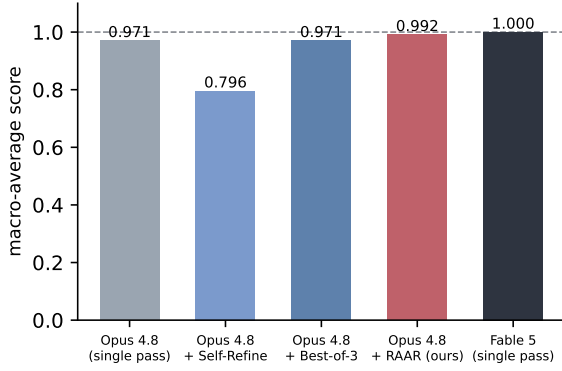


Fig. 2. Macro-average over the five objective tasks. The dashed line is the Fable 5 single-pass target.

4.8 single-pass already matches Claude Fable 5 at ceiling — code synthesis with adversarial edge cases, seeded bug fixing, and both novel combinatorics problems no longer separate these generations at this difficulty. The entire measured objective gap lives in T4, the 12-constraint precision-writing task, where Claude Fable 5 is exact (12/12 on all 4 runs) and Claude Opus 4.8 reliably miscounts. The failures are all exact-counting constraints — the sentence count, the total and first-sentence word counts, and the exactly-one-exclamation requirement (C1–C3, C8); the non-counting constraints (keyword frequencies, positional and length caps) hold. This specificity is itself a contribution of the contamination-controlled design: with novel, mechanically scored items, “the successor is better” reduces to a specific, measurable competence — exact composition under simultaneous counting constraints — rather than a diffuse impression.

(2) RAAR closes most of the gap that exists and opened no new one in any observed run. On T4, RAAR lifts Claude Opus 4.8 from 0.854 to 0.958, recovering 5 of the 7 constraint checks the base arm missed (71%, $n=4$ runs per arm with scores quantized in twelfths); on the macro-average it reaches 0.992 vs. Claude Fable 5’s 1.000. The single-task provenance matters: T4 is the only objective task with headroom, so the macro-average movement is the T4 movement, and at this n the difference between RAAR (0.958) and Self-Refine (0.979) on T4 is not separable from noise — the cross-task pattern, not any one cell, is the evidence. Equally important is the no-harm observation: across all 14 RAAR runs, not one scored below the Claude Opus 4.8 single-pass mean on its task (an all-of-14 observation, compatible at 95% confidence with a true per-run harm rate as high as $\sim 20\%$). The loop’s gate refused 8 of 14 first-round fused answers (Section VI-A), and no run that passed through revisions ended below its task’s base mean.

(3) Iteration without structure is actively destructive — replicating the negative literature. Self-Refine destroyed T3a: the base model answers 293,749 correctly in 3 of 3 single passes, but after two self-feedback rounds the final answers were 293,589, 220,815, and 293,875 — three

TABLE II
Blind pairwise judgments (wins out of 8; 4 judges $\times$ 2 position-swapped orders). 4–4 indicates no detectable preference.

comparison	T6 design	T7 explain
Opus 4.8 vs. Fable 5	0 – 8	0 – 8
RAAR vs. Fable 5	4 – 4	0 – 8
RAAR vs. Opus 4.8	8 – 0	6 – 2
RAAR vs. Self-Refine	4 – 4	1 – 7

different corruptions, scoring 0/3. The trajectory transcripts (shipped in the artifact) confirm at least one chain entered refinement with the correct answer and was talked out of it by its own critique (Section VI-C), the precise failure mode documented by Huang et al. [4] and the sharpening literature [9]. The same iteration budget inside RAAR’s structure (fresh-context adversarial verification, evidence-gated revision) caused zero such corruptions in 14 runs. Best-of-3, sharing RAAR’s sampling diversity but not its verification, tracked the baseline exactly (0.971) — diversity alone neither helps nor hurts here.

C. Semi-Objective: Seeded Code Review (T5)

All five arms recalled all six seeded defects (6/6) — defect-finding recall at this density does not separate these models either. One qualitative asymmetry survives: the module contains a genuine defect we did not plant (a negative-quantity order credits the buyer’s account). Claude Fable 5 and every augmented Claude Opus 4.8 arm (Self-Refine, best-of-3, RAAR) surfaced it; Claude Opus 4.8 single-pass alone did not. Consistent with the objective results, added inference structure recovered a successor-level finding that the bare predecessor pass missed, though with $n=1$ this is an observation, not a statistic.

D. Judged Open-Ended Tasks

Table II reports blind, position-swapped pairwise verdicts from the four judges (GPT-5.5 judge, Gemini, Claude Fable 5, Claude Opus 4.8; 8 verdicts per pair). Because the two orders are judged by the same judges, verdicts within a cell are correlated; we therefore report exact counts and unanimity rather than significance tests.

On the design task the panel can no longer distinguish the looped predecessor from the successor. Claude Opus 4.8 single-pass loses to Claude Fable 5 0–8 — every judge, both orders. RAAR beats its own base 8–0 with the same unanimity. Against Claude Fable 5 the count is 4–4, and the structure of the split is the finding: every one of the four judges voted for position A in both orders, so the verdicts became purely order-determined. We read this as indistinguishability rather than “parity” in any stronger sense — when the quality signal vanishes, position bias is all that remains, and that is precisely what unanimous position-following indicates. This is the configuration the kick predicts: a design document is a bundle of independently checkable commitments (algorithm choice, partition behavior, capacity arithmetic,

failure modes), exactly the structure an externalized rubric and an evidence-quoting verifier can grind through.

On the explanation task the gap narrows but does not close. RAAR beats its base 6–2, but Claude Fable 5 still wins 8–0 against both Claude Opus 4.8 arms. Notably, Self-Refine beats RAAR 7–1 here — the one setting in our study where unstructured polish outperforms structured verification. The pattern is informative: T7 quality is dominated by pedagogical coherence — one running example sustained across a long causal chain — which a holistic rewrite preserves and fusion-of-three-candidates fragments. Together with T3a (where Self-Refine destroys correctness that RAAR preserves), the two tasks bracket the design space: verification-shaped tasks reward RAAR; coherence-shaped tasks reward gentler iteration; and using the wrong loop for the task class is not a small mistake in either direction.

Two hygiene notes. First, output length: RAAR answers are the longest in every judged cell (T6: 12.7k characters vs. Claude Fable 5’s 9.2k and Claude Opus 4.8’s 8.5k; T7: 8.5k vs. 5.8k), so verbosity bias [30] plausibly assists RAAR in the cells it wins; we note, however, that the two cells RAAR loses unanimously (T7 vs. Claude Fable 5 and vs. the shorter Self-Refine) go against the length gradient, so length alone does not explain the pattern. Second, one judge verdict was unstable: a duplicated GPT-5.5-judge run on one T7 RAAR-vs.-base comparison returned opposite verdicts; we keep the first-run verdict (6–2; the alternative resolution gives 7–1, favoring RAAR more). The dedup rule — first verdict per (comparison, judge) — is fixed in the artifact.

Across both judged tasks, the predecessor single-pass takes 0 of 16 verdicts against the successor; with RAAR it takes 4 of 16, all on T6, where the panel’s preference signal disappears entirely.

V. Study 2: The Model-Tier Ladder on LvUP-Bench v2

A. LvUP-Bench v2 Benchmark Design

Study 1’s objective core largely saturated: four of five v1 objective tasks were already at ceiling for Claude Opus 4.8 single-pass. LvUP-Bench v2 was built to restore headroom for cheaper-tier comparisons. Its construction spec targets frontier single-pass scores between 0.3 and 0.85, requires pure text completion with deterministic scoring, authoritative prompts, novel instances, reference solutions that score 1.0, graded near-miss checks, and dual-computation for math ground truth. The resulting benchmark has 12 objective tasks and 3 judged tasks, but the post-hoc Fable results show that the frontier target was only partially achieved: Fable base is at or near ceiling on most scored objective tasks. We therefore use v2 for measured cheaper-tier headroom and report frontier saturation as a limitation, not as evidence of broad frontier discrimination. The validator certifies structural reproducibility, not publishability by itself: empirical difficulty, novelty, prompt ambiguity, and the semantic meaning of partial-credit magnitudes still require audit.

The primary objective set is 11 tasks across code (4), spec compliance (2), math (4), and seeded code review (1); the twelfth objective task, `constrained_writing_16`, is reported transparently but excluded from primary analysis because collection failed and left an incomplete no-answer cell. In the released artifact that task leaves a partial cell only: `base@sonnet` has one scored run at 0.625 and one failed run, while `base-direct@sonnet` and both `raar@sonnet` runs are failed no-answer cells, so the task does not support a fair paired comparison.

All v2 items are novel to this study. Code and spec-compliance tasks use hidden tests or exact file transformation; math tasks use exact answers verified by two independent programs; the code-review task uses seeded-defect recall; and the judged tail consists of `design_pipeline`, `explainRAFT`, and `migration_plan`. Collection is intentionally transparent about incompleteness. In the released scored artifact, the scored objective-row counts are `base@sonnet` $n=34$, `raar@sonnet` $n=33$, `selfrefine@sonnet` $n=17$, `bo10@sonnet` $n=16$, `raar-norubric@sonnet` $n=15$, `raar-approval@sonnet` $n=15$, `base-direct@sonnet` $n=11$, `base@haiku` $n=22$, `raar@haiku` $n=16$, `base@fable` $n=19$, and `raar@fable` $n=6$. We drop the Opus v2 cells from analysis because that batch is incomplete and the base point shows an unexplained anomaly.

For judged tasks, completed Sonnet-vs.-Sonnet comparisons contribute $n=2$ generation pairs, while frontier comparisons contribute $n=1$ pair when only one frontier generation completed. Study 1’s four-judge panel established that the Claude-family judges never flipped cross-family majorities, so v2 keeps only the cross-family pair (GPT-5.5 judge and Gemini), still with position swap, yielding 4 judgments per generation pair and a simpler conflict profile.

B. Model-Matrix Results

Figure 3 gives the scored-available ladder view, and Fig. 4 the same cells against mean calls. On the observed objective cells, the ladder rises from `base@haiku` (macro 0.376 over $n=11$ scored task means from 22 scored rows) to `base@sonnet` (0.721 over $n=12$ from 34 rows), while `base@fable` reaches 0.972 over $n=10$ from 19 rows. The looped and ablated Sonnet/Haiku/frontier arms are `raar@haiku` 0.650 ($n=11$ task means; 16 rows), `raar@sonnet` 0.820 ($n=11$; 33 rows), `selfrefine@sonnet` 0.873 ($n=12$; 17 rows; 0.901 on the same 11-task primary denominator used for the Sonnet-family comparison), `bo10@sonnet` 0.794 ($n=11$; 16 rows), `raar-norubric@sonnet` 0.834 ($n=11$; 15 rows), `raar-approval@sonnet` 0.841 ($n=11$; 15 rows), and `raar@fable` 0.967 ($n=5$; 6 rows). The direct-answer Sonnet control lands at 0.478 over 11 task means from 11 rows.

The scored-available view is descriptive; the inferential claims come from paired permutation tests on common tasks. On its full 11-task primary objective set, Haiku+RAAR gains 0.274 over Haiku base ($n=11$, $p=0.042$). Sonnet+RAAR gains 0.089 over Sonnet base

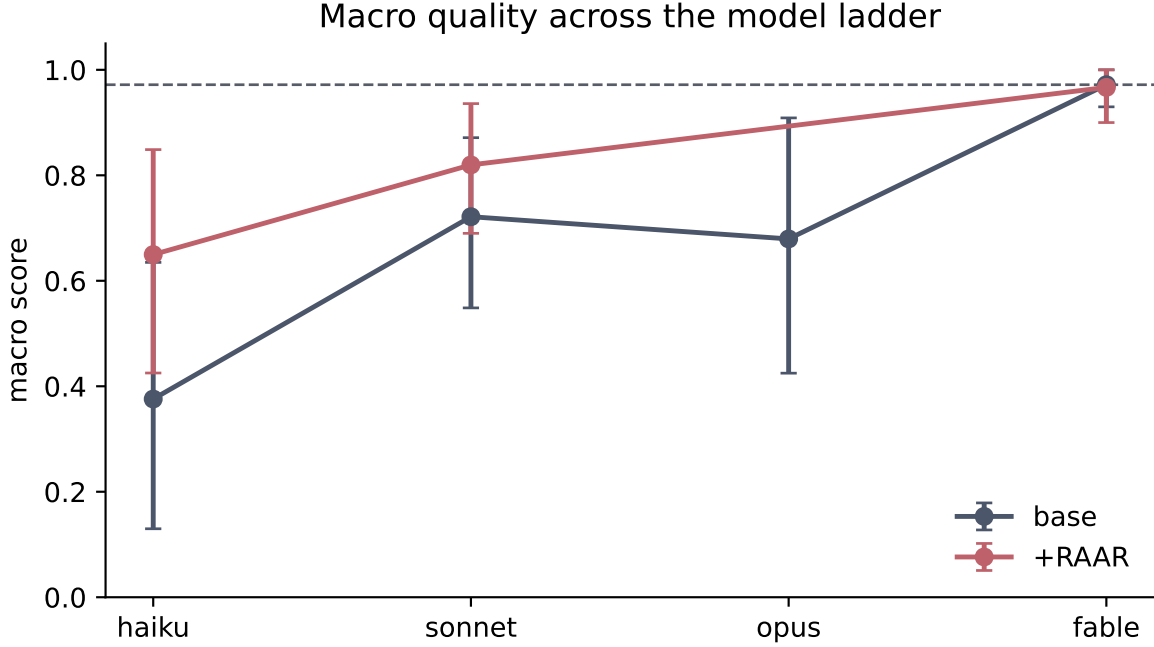


Fig. 3. Study 2 model ladder. Macros are computed over scored-available objective task means. Primary inferential claims in the text use common-task permutation tests and exclude `constrained_writing_16`.

($n=11$, $p=0.521$). Against Fable, however, the task-conditioned picture is sharper than the overall macro. Gap closure is computed only on tasks where Fable’s single-pass advantage over the base model exceeds 0.05 (the headroom rule fixed in `stats.json`); smaller denominators make the ratio unstable. Sonnet trails Fable by more than 0.05 on exactly 5 such common primary tasks — `absorb_expect`, `csv_normalize`, `digit_count`, `interval_sched`, and `seq_runs` — and Sonnet+RAAR closes 0.733 of that filtered gap (95% CI [0.300, 1.067], $n=5$). This number is fragile: the >0.05 cutoff excludes exactly `code_review_12` (denominator 0.042, a tiny-denominator ratio); at cutoff 0.0 or 0.02 closure falls to 0.500, at 0.10 it is 0.667, and pooled at the macro level over all 10 common tasks it is 0.419. We report the full sensitivity rather than a single point. Across all 10 common scored tasks, Sonnet+RAAR still sits 0.137 below Fable single-pass ($n=10$, $p=0.127$), so the point estimate still favors Fable even though the filtered headroom slice now excludes zero.

The Haiku result is more limited. Haiku+RAAR does improve in absolute score by 0.274 on the 11-task primary set, but its mean Fable-gap closure is only 0.304 (95% CI [-0.111, 0.639], $n=9$ tasks under the same >0.05 -headroom rule). The practical message is therefore asymmetric: one tier down can get close on the verification-shaped slice; two tiers down does not recover reliably.

C. Direct-Answer Control: Base-Direct vs. RAAR

The Sonnet direct-answer control remains mechanistically informative even though it no longer clears the 0.05 line. The base-direct ablation — a single-call answer-only

control without the role-separated scaffold — has macro 0.478 on 11 primary task means and trails Sonnet+RAAR by 0.341 on the 11 matched primary tasks ($p=0.061$). The conservative claim is therefore directional rather than confirmatory: whatever the rest of the Sonnet-tier ablations do, the multi-call scaffold is load-bearing in magnitude. The loop does not merely wrap a better prompt around the same direct answer; removing the staged generation, verification, and fusion structure is strongly destructive in observed score.

D. Sonnet-Tier Ablations

Study 2 also asks whether the distinctive Study 1 ingredient — the rubric-anchored adversarial gate — clearly survives at a more capable tier and a different task mix. At this n , the answer is no. Relative to Sonnet+RAAR, best-of-10 differs by -0.026 on 11 matched tasks ($p=0.928$), Self-Refine by $+0.081$ on 11 tasks ($p=0.444$), approval-exit RAAR by $+0.027$ on 10 tasks ($p=0.812$), and no-rubric RAAR by $+0.019$ on 10 tasks ($p=1.0$). With $n=2-3$ on the key Sonnet cells, Self-Refine is numerically the strongest Sonnet arm at macro 0.873, but the family of looped variants is statistically indistinguishable at this scale.

That negative result matters. Study 1’s objective pattern and transcript analysis pointed strongly at the gate as the loop’s kick. Study 2 does not clearly replicate that advantage at Sonnet tier on these task families. The shared Sonnet pattern is instead robustness: on the matched 11-task primary set, the loop family lifts base Sonnet from 0.730 to 0.820–0.901, i.e. about $+0.09$ to

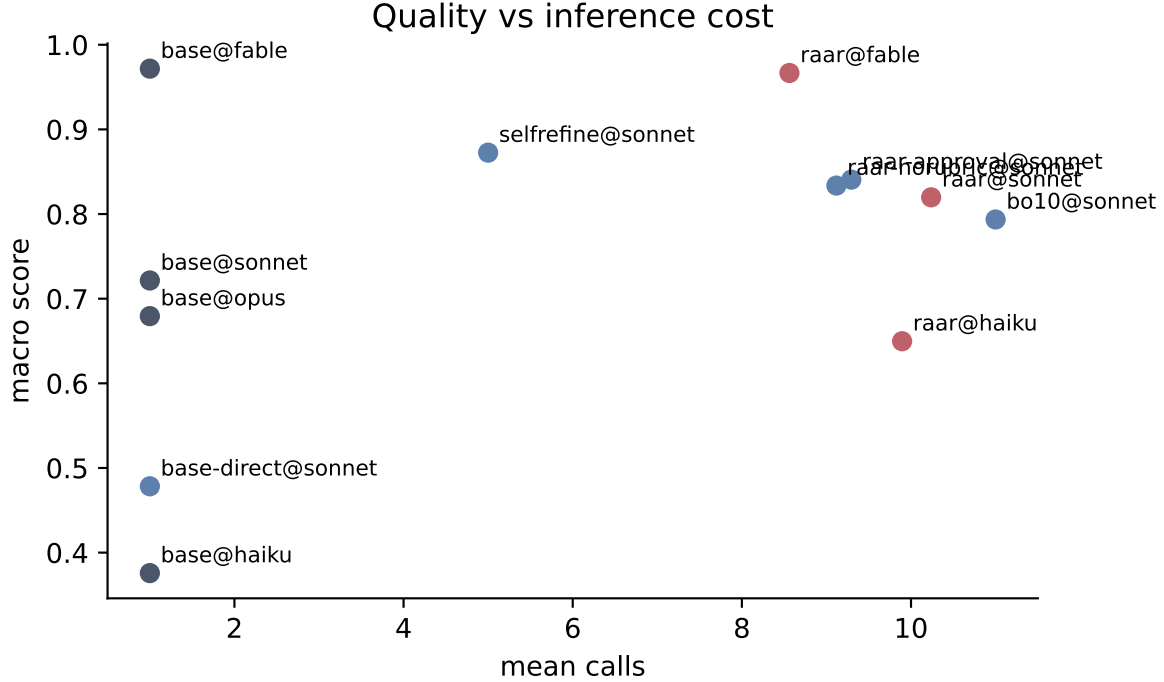


Fig. 4. Study 2 quality vs. inference cost. Each point is a scored-available objective cell; across all scored runs, Sonnet+RAAR averages 10.2 calls and Haiku+RAAR 9.9.

TABLE III

Study 2 judged outcomes. Each completed frontier cell is 4 judgments (2 judges $\times$ 2 answer orders); Sonnet-vs.-Sonnet cells use two generation pairs for 8 judgments. “-” marks a missing cell.

comparison	design	explain	migration
Sonnet base vs. Fable base	0 – 4	0 – 4	0 – 4
Fable +RAAR vs. Fable base	3 – 1	–	4 – 0
Sonnet +RAAR vs. Fable base	0 – 4	0 – 4	0 – 4
Sonnet +RAAR vs. Sonnet base	6 – 2	4 – 4	7 – 1

+0.17; the 0.901 endpoint is Self-Refine on that matched denominator, while its scored-available macro is 0.873 over 12 task means. The measured differences between sensible scaffolds compress into noise at this n . The honest reading is that the gate story is strongest when the base model still has visible headroom and the tasks are tightly verification-shaped.

E. Judged Open-Ended Tasks

Table III reports the judged tail. The judged artifact now contains 14 completed generation pairs, i.e. 28 blinded position-swapped comparisons evaluated by the two-family panel (GPT-5.5 judge and Gemini) for 56 total verdicts. Single-pair frontier comparisons contribute 4 judgments; the Sonnet-vs.-Sonnet cells use two generation pairs for 8. We do not run significance tests on these cells.

Fable base still sweeps Sonnet base 4–0 on all three judged tasks, and Sonnet+RAAR still loses 0–4 to Fable on all three. In other words, the open-ended gap is not closed even one tier down. Against its own base,

however, Sonnet+RAAR is sharply split by task family: it wins design_pipeline 6–2, ties explain_raft 4–4, and wins migration_plan 7–1. The new Fable+RAAR judged pairs are descriptive only but directionally favorable: 3–1 on design_pipeline, 4–0 on migration_plan, and no completed explain_raft pair. The coherence boundary remains where Study 1 already suggested it would be: migration planning is rich in independently checkable operational commitments, whereas the Raft explanation is a long causal exposition whose quality depends on coherent pedagogical flow.

F. Domain-Effectiveness Map

The numeric checker scores support the same task-shape story. Fig. 5 visualizes the Sonnet category deltas, while the domain map in Table IV summarizes docs/DOMAIN_MAP.md; by construction it uses numeric checker scores only, and buckets with no scored base/RAAR overlap are omitted. The table shows the Haiku and Sonnet tiers, where enough base/RAAR overlap exists for domain-level interpretation.

The practical playbook is therefore narrow and actionable. Use RAAR when the task is verification-shaped — with the clearest v2 support on math-like work — and when the underlying tier still has headroom. Code-review evidence is tiny-n and mixed, so we do not treat it as strong support. That verification-shaped region is where the detector-generator asymmetry appears most useful. Do not reach for it first on coherence-shaped prose, as the explain_raft loss and the Study 1 explanation task

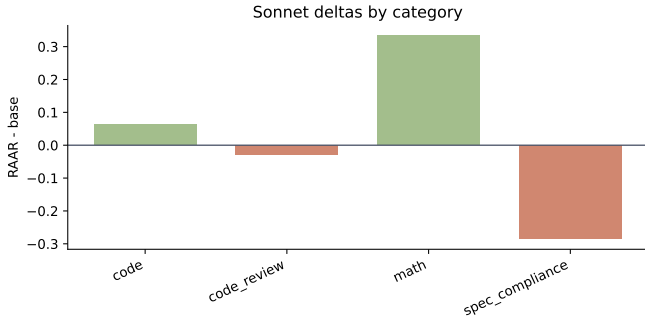


Fig. 5. Study 2 Sonnet deltas by category. Positive bars indicate RAAR helps the Sonnet tier on average within that category.

TABLE IV

Study 2 domain-effectiveness map from numeric checker scores only. Δ = RAAR - base within category; omitted cells had no scored base/RAAR overlap.

category	verif.	Haiku Δ	Sonnet Δ	verdict
code	high	-0.012 ($n=4$)	+0.062 ($n=4$)	neutral
code review	med.	+0.083 ($n=1$)	-0.028 ($n=1$)	mixed helps
math	high	+0.625 ($n=4$)	+0.333 ($n=4$)	strongly
spec compl.	high	+0.238 ($n=2$)	-0.286 ($n=2$)	neutral

both warn. Be skeptical on spec-compliance tasks once the tier is already capable: effects there are mixed and near zero. And do not replace the role-separated scaffold with a bare direct answer: the base-direct ablation is the largest Sonnet-tier drop even though it no longer separates cleanly from noise.

G. Practical Numbers and Incomplete Cells

For readers facing usage-based pricing, the headline numbers are straightforward. One tier down plus RAAR gets surprisingly close on the part of the distribution that is rich in checkable structure: Sonnet+RAAR closes about 73% of the filtered Fable gap on the 5 common primary tasks where Fable’s headroom over Sonnet exceeds 0.05, yet remains 0.137 behind Fable over all 10 common scored tasks ($p=0.127$). The cost is roughly a ten-call cheaper-tier budget: across all scored runs, the collected Sonnet+RAAR runs average 10.2 calls, and the Haiku+RAAR runs 9.9. Two tiers down does not recover: Haiku+RAAR lifts absolute quality, but its mean gap closure is only 0.304 with a confidence interval that crosses zero.

Lost cells are part of the result, not bookkeeping noise. The released JSON contains failed-run no-answer rows for the still-incomplete equal-scaffolding frontier matrix, and the expanded Fable-base probes add failed no-answer single-pass rows for perm_constraints and seq_runs. Those salvaged attempts yielded no scorable answer and are treated as collection failures, not as model-quality scores. The artifact is explicit about what completed, what failed, and what was excluded.

Does the loop lift the frontier model itself?: An earlier Fable+RAAR batch was interrupted mid-collection, but the expanded artifact now contains a narrow completed frontier slice. On six common scored objective cells spanning five verification-shaped tasks, raar@fable averages 0.967 versus base@fable at 0.951 on the same five task means (base@fable is 0.972 over its broader 10-task scored-available macro), with 0 degraded runs out of 6 compared cells and paired-permutation $p=0.498$ for the +0.015 common-task difference. On this slice the loop neither lifts nor harms the frontier tier measurably.

VI. Study 1 Mechanism Checks

The transcript-level analyses below remain specific to Study 1, which is the only study with the full predecessor/successor loop comparison and the full sanitized role-separated message transcripts needed to diagnose mechanism.

A. Gate Behavior

Across the 14 objective RAAR runs the gate issued 22 verdicts: 9 PASS, 13 FAIL. Eight of 14 first-round fused answers were refused — the adversarial mandate demonstrably does not rubber-stamp. By task: both T1 and both T2 runs passed the first gate; all four T4 runs and two of three T3a runs drew first-gate refusals; T3b runs were refused twice and passed once on the first gate. Five runs (36%) exhausted the $R=2$ revision cap still under verdict FAIL — and, per Algorithm 1, a cap-exhausted run returns its final revision unverified. In our data those five unverified finals scored well (a T4 run that left the gate at FAIL/FAIL scored 12/12; both cap-exhausted T3a runs ended exactly correct, including the poisoned-rubric run), so in no observed run did the gate’s false-FAIL direction cost correctness rather than compute — but this is an empirical observation about 5 runs, not a property the design guarantees. The converse error matters more: in the two T4 runs with a residual defect, it was the same constraint both times — the exact first-sentence word count. Counting is the one faculty where detector and generator share the same weakness, so the geometric-suppression argument of Section III-D loses its decorrelation premise; this is the boundary of the method, and we report it as such.

B. Case Study: a Poisoned Rubric Cannot Derail the Loop

The sharpest stress test of RAAR arrived uninvited. In one T3a run the Stage-1 rubric instance violated its no-solving instruction: its first criterion declared “the response’s final count equals the true value 397,530” — a number it had derived itself, incorrectly. The wrong target was thereby baked into the grading contract that every downstream verifier inherited, the worst-case failure of rubric externalization. The gate duly FAILED the (correct) fused answer of 293,749 twice. The loop survived anyway: because revision instances are instructed to re-verify every

flagged defect rather than obey the report, the reviser re-derived the count, rejected the rubric’s figure, and the run exhausted its cap still holding 293,749 — scoring 1.0. Robustness to wrong critiques — even a wrong rubric — is a direct consequence of burden-flipping: in RAAR, critiques and criteria are claims to be checked, never instructions to be applied. The episode also tempers the Stage-1 design claim: rubric exogeneity is an instructed tendency, not a guarantee, and the loop’s safety rests on the re-verification discipline beneath it.

C. Case Study: How Self-Refine Destroys a Correct Answer

The shipped T3a Self-Refine transcripts show the corruption mechanism in full. A draft entered refinement holding the correct 293,749. The self-critique conceded the answer’s format was correct, then applied an independence heuristic of its own invention — multiplying the unconstrained count by 0.9^6 for the adjacency constraints — to declare a “plausible band” of roughly 215,000–235,000, flagged the correct answer as “noticeably above that band,” and ordered the refinement to “not trust 293,749 — recompute from scratch.” The hand recomputation then slipped, as hand recomputations do; one run’s final answer, 220,815, landed inside the wrong heuristic’s band — the model refined its answer toward its own critique’s mistaken expectation. Three independent trajectories produced three different wrong answers, the instability signature of critique uncorrelated with truth. The structural diagnosis: the critic shares the generator’s context and commitments, so its objections sample from the same error distribution as the generation, and iterating that loop performs a random walk around the answer. RAAR’s verifiers are instead anchored to per-criterion evidence obligations and mandatory recomputation — and its revisers are licensed to reject the critique itself, which is what saved the poisoned-rubric run above.

D. Cost

RAAR spends 9–12 base-model calls (10.5 mean) against a single-pass target’s 1; stages 2 and 3 parallelize, so the critical path is 5–7 sequential calls. Whether the trade is favorable is a deployment question that depends on the price ratio between generations at the time of use, and we make no economic claim. The scientific claim is orthogonal to price: the measured quality difference is, on these tasks and under these bounded conditions, partly recoverable from the base weights by structure alone.

VII. Code, Data, and Reproducibility

The companion artifact is organized so that each claim can be traced back to source prompts, raw outputs, scored rows, judge manifests, transcripts where available, and analysis scripts. Study 1 task prompts and checkers are under `bench/tasks/` and `bench/checkers/`, with scored objective rows in `results/all_objective.scored.json`. Sanitized

Study 1 role-separated message transcripts are under `results/transcripts/study1_math_batch/`; they preserve ordered user/assistant message content while removing local workflow metadata. Study 2 task prompts are the prompt fields in `bench/v2/tasks/*/task.json`; the structural benchmark validator is `bench/v2/validate.py`; raw and scored runs are under `results/v2/`; and the statistics used in the paper are produced from `results/v2/all_runs.scored.json` by `analysis/v2_stats.py`. The RAAR role prompts, rubric prompt, verification prompt, fusion prompt, revision prompt, and ablation arms are in `harness/lvup.py` and `harness/lvup_runner_v2.js`. Blinded judged-task prompts, manifests, and verdicts are under `results/judging/` and `results/v2/judging/`.

The intended reproduction path is: run `bench/v2/validate.py` to check the benchmark package; run `analysis/v2_stats.py` on `results/v2/all_runs.scored.json` to regenerate `results/v2/stats.json` and the printed tables; run `analysis/v2_domain_map.py` for the domain-effectiveness summary; run `analysis/v2_judging/aggregate_judgments.py` for judged-task aggregates; and rebuild the PDF from `paper/main.tex` with the five included figure PDFs. The paper reports completion-conditional scored means and separately describes missing/no-answer cells; it does not impute failed collections as model scores.

VIII. Limitations

Adversarial audit. An independent adversarial reviewer (a separate model family, instructed to break the central claim) raised three issues we treat as load-bearing. (i) Benchmark hardening: the four v2 code-task checkers originally let candidate code reach the shipped hidden fixtures, so a fixture-reading cheat submission could score perfectly. No evaluated model exploited this — all submissions are genuine solutions, so the reported scores are unchanged — and the released artifact ships a hardened-checker patch that runs candidates in an isolated working directory with the fixtures inaccessible; the benchmark should be treated as contamination-proof for reuse only with that patch applied. (ii) Headline fragility: the 0.733 Sonnet gap-closure figure depends on the >0.05 tiny-denominator headroom cutoff and ranges over 0.50–0.73 (0.42 pooled); we now report the full sensitivity. (iii) No equal-scaffolding frontier control: the headline compares Sonnet+RAAR against unscaffolded Fable, not Fable+RAAR; our partial Fable+RAAR slice (0.967 vs. 0.951 base on five common tasks, $+0.015$, $p=0.498$; base@fable is 0.972 over the broader 10-task scored macro) is consistent with no measurable frontier lift but does not establish equal-scaffolding parity. The honest surviving claim is therefore the narrow one: on a filtered verification-shaped subset, one tier down plus RAAR recovers roughly half to three-quarters of the gap to unscaffolded frontier output; the stronger “approaches the frontier” reading is not established by these data.

Scale and incompleteness. Both studies are intentionally small. Study 1 has 8 tasks and 2–4 runs per objective

cell; Study 2 has 15 tasks overall, but primary objective inference rests on 11 tasks, and judged comparisons on 1–2 generations per arm. Several Study 2 cells are explicitly missing as failed no-answer runs: the entire Opus+RAAR objective batch and two Fable single-pass cells are recorded as failed no-answer collections; the completed Fable+RAAR batch covers 5 common objective tasks and 2 judged tasks. The large effects are informative; small differences are not.

Primary-analysis exclusions. `constrained_writing_16` is part of the v2 benchmark but excluded from primary analysis because collection failed and left an unbalanced no-answer partial cell. Some Study 2 figures therefore show scored-available summaries rather than a perfectly balanced matrix; the inferential claims always come from the common-task permutation tests in `stats.json`.

No content-filter attribution. We do not attribute failed no-answer rows, missing cells, or RAAR gate refusals to provider content filters or moderation events. The artifacts do not record such provider-side metadata, so these rows are treated only as failed/no-answer collection artifacts and not as evidence about safety filtering.

Checker granularity. v2’s near-miss files prove that checkers can distinguish plausible flawed answers from references, but they are not semantic calibrations of error severity. Some exact-output transformation tasks use line- or case-level partial credit, so their numeric partial scores should be read as deterministic task scores rather than calibrated human utility.

Missing and weak ablations. Study 1 points strongly at the gate, but Study 2 does not clearly replicate that advantage at Sonnet tier. The available Sonnet ablations are still thin, partially missing, and mostly single-run outside the expanded key cells. Approval-exit, no-rubric, best-of-10, and Self-Refine are therefore useful negative evidence, not clean component isolations. Fully call-matched, fully completed component ablations remain future work.

Asymmetric scaffolding by construction. Study 1 asks what is recoverable from older weights against a successor as deployed, so only the predecessor is scaffolded. Study 2 asks the practical cheaper-tier question, but the full-ladder equal-scaffolding matrix remains incomplete even after the added Fable+RAAR cells, so the frontier-loop claim is limited to a narrow verification-shaped slice.

No GPT-5.5 generation arm. The v2 scored artifact contains generation rows for Haiku, Sonnet, Opus, and Fable only. GPT-5.5 appears in the judging artifacts, not in `all_runs.scored.json` as generated answers. A valid GPT-5.5 generator comparison would require a new, same-protocol collection pass with the same no-tools setting, base/direct and RAAR arms, scoring scripts, and missing-cell accounting. Until then, GPT-5.5 should be treated as a judge in this paper, and as a proposed generator extension rather than a generator result.

Task-shape boundaries. The combined result is not that RAAR dominates. Study 1’s explanation task and Study 2’s `explain_raft` cell both show a coherence-shaped weakness, while exact counting leaves a residual gap even

in Study 1. Tool grounding would likely change those boundaries; we exclude tools here to keep the claim purely about call structure.

Judge validity. Open-ended results still rest on LLM judges. We mitigated the addressable biases with blind anonymization and position swaps, and Study 2 simplified the panel after Study 1 showed that the Claude judges never flipped cross-family majorities. Even so, judged cells are few, correlated, and only a proxy for expert human evaluation.

Inference-setting opacity. Arms ran at vendor-default settings via the Claude Code subagent interface; provider-side configuration (e.g. reasoning defaults) is not independently controllable by us. The comparison is therefore “deployed default vs. deployed default,” which matches the practical question but not a parameter-matched laboratory ideal.

IX. Conclusion

These two studies answer two different deployment questions with the same loop. Study 1 shows that a direct cross-generation gap can be materially recovered from older weights alone: on LvUP-Bench v1, RAAR recovers 5 of the 7 constraint checks that constitute the entire objective gap (71%), reaching 0.992 vs. 1.000 macro-average with no degraded run observed in 14. Study 2 broadens the claim from predecessor vs. successor to cheaper tier vs. frontier tier: on LvUP-Bench v2, one tier down plus RAAR closes 0.50–0.73 (threshold-dependent; 0.42 pooled) of the filtered gap to unscaffolded Fable on the verification-shaped tasks where Sonnet still trails Fable by more than 0.05 under the documented headroom rule — not an equal-scaffolding frontier-equivalence result, since the Fable+RAAR control is incomplete — but two tiers down does not recover, and the open-ended coherence gap remains.

The mechanism is not merely more calls, but differently structured inference. Study 1 already showed that best-of-3 changes nothing while Self-Refine can catastrophically destroy a correct reasoning answer; the v2 direct-answer control sharpens that diagnosis by showing that removing the role-separated scaffold drops Sonnet from 0.820 to 0.478 (matched-task $\Delta = -0.341$, $p=0.061$). At the same time, Study 2 tempers any stronger claim: the specific gate advantage that looks salient in Study 1 does not clearly replicate at Sonnet tier, Self-Refine is numerically higher than full RAAR on the matched Sonnet slice (0.901 vs. 0.820) but not statistically separable at this n , and the judged open-ended gap is not closed even one tier down.

The practical reading is therefore precise rather than universal. If the task is verification-shaped and the cheaper tier still has headroom, RAAR can buy back a meaningful share of the gap to unscaffolded frontier output at roughly a ten-call budget. If the task is coherence-shaped prose, already-near-ceiling spec compliance, or a two-tier drop, the loop is not enough. The scientific reading is the same in stricter language: some capabilities measured by a generation or tier gap are latent in the weaker model and

extractable by orchestration alone, but the recoverable share is sharply bounded by task shape, model tier, and the quality of the model’s own verification.

That boundary is the main product of the paper. The artifact is meant to give future work concrete handles: complete the equal-scaffolding frontier matrix, add non-Claude generator families, replace LLM-judged tails with human or expert panels, and isolate rubric, verifier, fusion, and gate components under call-matched controls. If those extensions move the boundary, the present paper should make it clear which part moved and why.

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